

# Prosun Kanti Datta

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## Technical Stack

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- **Programming:** Python (NumPy, Pandas, SciPy, Matplotlib, PyTorch)
- **Scientific Computing:** Monte Carlo simulation, introductory Machine Learning, basic Physics-Informed Neural Networks (PINNs)
- **Tools:** Git, GitHub, VS Code, Vim, MS Excel, MS Word, LaTeX
- **Systems & Web:** Linux (Ubuntu/Debian), Windows, HTML, CSS

## Education

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- **Higher Secondary Certificate (HSC), Science Group** Expected 2027  
British Bangladesh International School & College, Sylhet (English Version)
- **Secondary School Certificate (SSC), Science Group** Completed 2025  
British Bangladesh International School & College, Sylhet (English Version)  
GPA: **5.00 / 5.00**

## Research Interests

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- Computational Physics
- Scientific Machine Learning
- Atomic & Molecular Dynamics
- Stochastic Processes & Uncertainty Quantification
- PIML for Renewable Energy Systems
- Energy Systems Modeling

## Research Experience

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**Lead Researcher** *Ongoing*  
Physics-Informed ML for Solar Microgrid Resilience in the Sylhet Monsoon Corridor

- Leading a student-initiated interdisciplinary research project on solar microgrid resilience in surgical facilities under monsoon-driven uncertainty.
- Responsible for physics-informed modeling and machine learning components, with focus on stochastic analysis, system resilience, and techno-economic evaluation.

**Independent Researcher** *Completed*  
Stochastic Modeling of Radioactive Decay via Monte Carlo Simulation  
 [GitHub Repository](#)

- Developed a Python-based Monte Carlo simulation to model radioactive decay as a stochastic physical process.
- Explored probabilistic decay behavior to strengthen understanding of computational physics and numerical simulation.

## Achievements

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**National Finalist** 2025  
Bangladesh Olympiad Challenge (BDOC) – Recognized as a finalist in the national Math Olympiad,

demonstrating strong quantitative reasoning and analytical problem-solving.

 [Certificate](#)

**Captain, School Football Team**

2024–Present

Led the team as captain and striker to 2 championship titles and 1 runner-up finish, reflecting leadership, sustained performance, and high-level teamwork.